

38. Case Study 3: The Brooch (Areas Related to Circles)

A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in the figure.

Based on the above information, answer the following questions:

- (i) Find the total length of the silver wire required. (1 Mark)
- (ii) What is the angle of each sector of the brooch? (1 Mark)
- (iii) **(a)** Find the area of each sector of the brooch. (2 Marks)

OR

(b) If the diameter was doubled, what would be the area of each sector? (2 Marks)